

A bisection identity between OEIS A111403 and OEIS A002716

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Abstract

OEIS A111403 carries a conjecture of R. J. Mathar that its n -th term is one less than the $(2n)$ -th term of OEIS A002716, Somos's infinite coprime sequence. It is true. The recurrence A002716 states as fact splits into two closed forms, one for each parity of the index, and a short induction proves both at once: the odd-indexed terms are exactly the Fermat numbers $2^{2^{n+1}} + 1$, and the even-indexed terms are $2^{2^{n+1}} - 2^{2^n} + 1$, which is one more than the defining expression for A111403.

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1 The two sequences and the conjecture

OEIS A111403 is “ $a(n) = f(f(n+1)) - f(f(n))$, where $f(m) = 2^m$.”. It has offset 0 and begins

2, 12, 240, 65280, 4294901760, 18446744069414584320, 3402823669209384634444927863358058659840, ...

OEIS A002716 is “An infinite coprime sequence defined by recursion.”. It has offset 0 and begins

3, 5, 13, 17, 241, 257, 65281, 65537, ...

The entry A111403 carries the following comment:

Conjecture: $a(n) = A002716(2*n) - 1$. — *R. J. Mathar*, May 15 2007

As of the “Last modified” line on the live entry (Oct 06 2024 09:17:07, revision 29) the statement is still recorded as unproved, and nothing on either entry records it as settled either way. (A002716 carries a separate conjecture of the same contributor, on its odd-indexed terms; it is not the statement treated here, and it is not used below.)

Write $f(m) = 2^m$. The name of A111403 defines

$$a(n) = f(f(n+1)) - f(f(n)) = 2^{2^{n+1}} - 2^{2^n}, \quad (1)$$

and the statement to be proved is

$$a(n) = b(2n) - 1 \quad \text{for all } n \geq 0, \quad (2)$$

where b is A002716.

2 What A002716 states as fact

A002716 states, not as a conjecture,

$$a(2*n + 1) = a(2*n) + a(2*n - 1) - 1, \quad a(2*n) = a(2*n - 1)^2 - 3 * a(2*n - 1) + 3, \quad a(0) = 3, \\ a(1) = 5.$$

that is, with b for the sequence,

$$b(0) = 3, \quad b(1) = 5, \quad b(2n) = b(2n-1)^2 - 3b(2n-1) + 3, \quad b(2n+1) = b(2n) + b(2n-1) - 1 \quad (3)$$

for $n \geq 1$. This determines b completely, and it is the only input taken from that entry; it was checked against all 13 of its published terms before use.

3 The proof

Lemma 1. *For every $n \geq 0$,*

$$b(2n+1) = 2^{2^{n+1}} + 1,$$

and for every $n \geq 1$,

$$b(2n) = 2^{2^{n+1}} - 2^{2^n} + 1.$$

Proof. Induction on n , proving the odd-index statement and using it to get the next even-index one.

For $n = 0$ the first claim reads $b(1) = 2^2 + 1 = 5$, which is the initial condition in (3).

Let $n \geq 1$ and suppose $b(2n-1) = 2^{2^n} + 1$, which is the first claim at $n-1$. Write $u = 2^{2^n}$, so $b(2n-1) = u + 1$ and $u^2 = 2^{2^{n+1}}$. By (3),

$$b(2n) = (u+1)^2 - 3(u+1) + 3 = u^2 + 2u + 1 - 3u - 3 + 3 = u^2 - u + 1 = 2^{2^{n+1}} - 2^{2^n} + 1,$$

which is the second claim at n . Then, again by (3),

$$b(2n+1) = b(2n) + b(2n-1) - 1 = (u^2 - u + 1) + (u + 1) - 1 = u^2 + 1 = 2^{2^{n+1}} + 1,$$

which is the first claim at n . The induction is complete. $\square$

Theorem 2. *For every integer $n \geq 0$, $a(n) = b(2n) - 1$, where a is OEIS A111403 and b is OEIS A002716. This is the statement of Section 1.*

Proof. For $n \geq 1$, Lemma 1 gives $b(2n) - 1 = 2^{2^{n+1}} - 2^{2^n}$, which is $a(n)$ by (1).

For $n = 0$ we have $b(0) = 3$ from (3), so $b(0) - 1 = 2$, while (1) gives $a(0) = 2^2 - 2^1 = 2$. The two agree. $\square$

Corollary 3. *The odd-indexed terms of A002716 are the Fermat numbers: $b(2n+1) = F_{n+1}$, where $F_m = 2^{2^m} + 1$.*

Proof. This is the first statement of Lemma 1. $\square$

4 Verification

Three checks, each independent of the algebra above.

First, (1) was evaluated in exact integer arithmetic and compared against every one of the 8 published terms of A111403; the values agree exactly.

Second, the recurrence (3) was iterated in exact integer arithmetic and compared against every one of the 13 published terms of A002716; the values agree exactly. The two closed forms of Lemma 1 were then evaluated independently and matched against the same terms at every index of each parity.

Third, the identity (2) was checked directly on the published terms, without any closed form: for every n with $0 \leq n \leq 5$ — the range for which both $a(n)$ and $b(2n)$ are published — the two integers agree.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026.
- [2] G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*, 6th ed., Oxford University Press, 2008.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011.